

Big Data Dataset

Data-Generation Scripts

The data is generated using a seeded simulator from src/utiq/simulate/ (network.py, operations.py, defects.py, ceiling.py). Command to run is:

```
utiq generate
```

This creates around 4.3 million rows in 12 tables (about 600 MB with both Parquet and final processed data). An initial smaller scale run can be done using UTIQ_SCALE=0.01 utiq generate.

Data Dictionary

Table	Rows	Columns	Primary Key	Partitioned By
stops	520	7	stop_id	-
routes	100	10	route_id	-
route_stops	3,068	5	route_id, direction, stop_order	-
vehicles	260	5	vehicle_id	-
service_calendar	365	4	service_date	-
schedules	900	7	service_id	-
passengers	50,000	4	passenger_id	-
trips	567,248	14	trip_id	month
passenger_counts	567,248	13	trip_id	month
delays	305,537	12	delay_id	month
tickets	2,012,000	10	ticket_id	month
gps_events	800,344	6	-	-

Dataset Schemas

All the 12 tables are declared explicitly in src/utiq/schemas.py , a central registry of tables, instead of being inferred at read time. The ingestion process does a declared vs. inferred comparison to illustrate the reason why inference cannot be used because an integer column containing one null will become double on inference.

Primary / Foreign Key Definitions

Fifteen foreign key declarations tie 12 tables together. Every such declaration generates two automated checks for referential integrity and uniqueness during the quality.

Child Table	Column	Parent Table
route_stops	route_id	routes
route_stops	stop_id	stops
schedules	route_id	routes
trips	route_id	routes
trips	vehicle_id	vehicles
passenger_counts	trip_id	trips
passenger_counts	route_id	routes
delays	trip_id	trips
delays	route_id	routes
tickets	passenger_id	passengers
tickets	trip_id	trips
tickets	route_id	routes
tickets	board_stop	stops
tickets	alight_stop	stops
gps_events	trip_id	trips

Raw / Cleaned / Parquet Data

- Sample raw data: CSV output of utiq generate prior to any data transformations
- Cleaned data: utiq run clean output – data is either corrected, quarantined or deduplicated according to the audit trail
- Parquet data: utiq run ingest and utiq run integrate outputs – partitioned monthly on the four growing tables (trips, passenger counts, delays, tickets)

Training / Validation / Test Data

All of the models used in this project use the exact same trip_features table, sliced across the exact same two dates (2025-09-01 and 2025-10-01), from the exact same configuration file:

Split	Months	Trips	Used For
Train	1–8	347,772	Fitting
Validation	9	51,267	Model / algorithm selection
Test	10–12	156,852	Reported results (scored once)

Dataset Statistics

Complete statistical analysis (range, mean, standard deviation, unique count by exact counting and not by approximation, and distribution of values) is created by utiq run catalog into data/reports/dataset_statistics.json and data_dictionary.json.